Group-01

Project Report:

Title: Brain Cancer Classification Using Deep Learning

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Abstract—

Brain cancer remains one of the most lethal malignancies worldwide, demanding accurate and early automated diagnosis. This paper presents a comprehensive comparative study of deep learning models for multi-class brain cancer classification using MRI scan images. We evaluate three transfer-learned architectures — ResNet-50, VGG-16, and EfficientNet-B0 — alongside a custom Convolutional Neural Network (CNN) augmented with a Convolutional Block Attention Module (CBAM). The dataset comprises 15,000 MRI images equally distributed across three brain cancer classes: Glioma, Meningioma, and Brain Tumor. Extensive data augmentation, 5-fold stratified cross-validation, and early stopping are applied to ensure robust generalization. Statistical significance of model differences is verified via paired t-tests (alpha = 0.05). The proposed CBAM-enhanced custom CNN, trained with cross-validation, demonstrates competitive performance while offering interpretable attention maps through Gradient-weighted Class Activation Mapping (Grad-CAM). Results indicate that attention mechanisms provide meaningful diagnostic focus, particularly for distinguishing glioma from meningioma.

Index Terms—Brain Cancer, MRI Classification, CBAM Attention, Transfer Learning, ResNet-50, VGG-16, EfficientNet-B0, Custom CNN, Grad-CAM, Paired T-Test, Deep Learning.

INTRODUCTION

Brain cancer is among the most aggressive and life-threatening forms of cancer, with gliomas, meningiomas, and other brain tumors collectively contributing to a significant proportion of cancer-related mortality. According to the World Health Organization (WHO), brain and central nervous system tumors account for approximately 308,000 new diagnoses globally each year [1]. Accurate and early diagnosis is critical, as treatment outcomes are strongly correlated with the stage at which the disease is detected.

Magnetic Resonance Imaging (MRI) is the gold standard for brain tumor identification and delineation. However, manual interpretation of MRI scans is time-consuming, subject to inter-observer variability, and demands specialized radiological expertise. The exponential growth of medical imaging data has made automated, AI-assisted diagnosis both feasible and necessary. Deep learning methods, particularly Convolutional Neural Networks (CNNs), have demonstrated remarkable success in image recognition tasks and hold tremendous promise for medical image analysis [2].

Transfer learning from large-scale models pretrained on ImageNet has become a dominant paradigm in medical imaging due to the limited availability of labeled clinical data. Architectures such as ResNet-50 [3], VGG-16 [4], and EfficientNet-B0 [5] have been widely adopted for this purpose. Simultaneously, attention mechanisms — particularly the Convolutional Block Attention Module (CBAM) [6] — have emerged as powerful tools to guide the network to focus on diagnostically relevant image regions, improving both accuracy and interpretability.

This paper makes the following contributions: (1) A systematic exploratory data analysis (EDA) of the multi-class brain cancer MRI dataset; (2) Fine-tuning and comparison of three state-of-the-art pretrained models under identical conditions; (3) Design and evaluation of a custom CNN integrated with CBAM attention and trained using 5-fold stratified cross-validation; (4) Statistical significance analysis of model differences using the paired t-test; and (5) Visual interpretability analysis via Grad-CAM heat maps.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and methodology. Section IV presents the CBAM attention diagram and architecture. Section V reports experimental results. Section VI concludes with future directions.

RELATED WORK

A. Deep Learning for Brain Tumor Detection

Deep learning has been extensively applied to brain tumor classification from MRI images. Abiwinanda et al. [7] proposed a simple CNN achieving 84.19% accuracy on a three-class brain tumor dataset. Sultan et al. [8] demonstrated that deeper CNN architectures with batch normalization significantly outperform shallow networks. Khan et al. reported that pretrained models fine-tuned on domain-specific data consistently surpass models trained from scratch on small datasets, underscoring the importance of transfer learning in the medical domain [2].

B. Transfer Learning in Medical Imaging

He et al. introduced the Residual Network (ResNet) [3], which addressed the vanishing gradient problem through skip connections, enabling training of very deep networks. ResNet-50, a 50-layer variant, has become a standard baseline in medical mage classification. Simonyan and Zisserman's VGG-16 [4], characterized by its uniform 3×3 convolutional filters arranged in 16 weight layers, has been widely adopted due to its simplicity and strong representational capacity. Tan and Le proposed EfficientNet [5], which scales network depth, width, and resolution jointly using a compound coefficient, achieving superior accuracy with fewer parameters than prior architectures.

C. Attention Mechanisms

Attention mechanisms allow neural networks to selectively emphasize informative regions of feature maps. Woo et al. introduced CBAM [6], a lightweight attention module that sequentially applies channel attention — identifying which feature channels are most relevant — followed by spatial attention — determining which spatial locations carry the most discriminative information. CBAM has demonstrated consistent improvements across multiple benchmark tasks with negligible computational overhead. In the medical imaging context, CBAM has been applied to skin lesion detection, retinal disease classification, and lung nodule segmentation with notable performance gains [9].

D. Model Interpretability

Interpretability is of paramount importance in clinical AI applications. Selvaraju et al. proposed Gradient-weighted Class Activation Mapping (Grad-CAM) [10], which uses the gradients flowing into the final convolutional layer to produce a coarse localization map highlighting important regions for a prediction. Grad-CAM has since been adopted as a standard explainability tool in medical imaging pipelines, enabling radiologists to validate model attention against known pathological markers.

E. Statistical Validation in Deep Learning

Dietterich [11] established foundational guidelines for statistical comparisons of machine learning classifiers, recommending the paired t-test for comparing per-sample correctness vectors across methods. This approach has been applied in medical AI benchmarking to assess whether observed performance differences between models are statistically significant rather than attributable to random variation, providing scientific rigor to model selection decision

METHODOLOGY

Dataset

This study uses a dataset consisting of 15,000 brain scan images distributed equally across three classes, namely brain glioma, brain meningioma, and brain tumor, with 5,000 images per class. All experiments were conducted on a GPU accelerated environment using PyTorch as the primary deep learning framework**.**

Data Preprocessing and Augmentation

The preprocessing pipeline was designed to prepare images in a consistent and generalizable manner. All images were resized to 224×224 pixels before being fed into the model. During training, a series of data augmentation techniques were applied including random horizontal flipping, random rotation up to 15 degrees, color jitter adjustments to brightness and contrast, and random affine translations. These augmentations were applied exclusively to the training data to artificially increase the diversity of the training set and reduce the risk of overfitting. Validation and test images were only resized and normalized without any augmentation, ensuring a fair and unbiased evaluation. All images were normalized using standard ImageNet mean and standard deviation values.

Model Architecture

The architecture of the custom CNN consists of five convolutional blocks stacked sequentially. Each block follows the pattern of Conv2d followed by Batch Normalization, ReLU activation, another Conv2d, Batch Normalization, ReLU activation, and finally MaxPooling to reduce the spatial dimensions by half. The number of feature channels doubles progressively across the five blocks, starting from 32 channels in the first block and reaching 512 channels in the fifth block. This design allows the network to learn increasingly abstract and complex visual features, beginning with low level patterns such as edges and textures in the earlier blocks and progressing to high level structural features in the deeper blocks.

Attention Mechanism

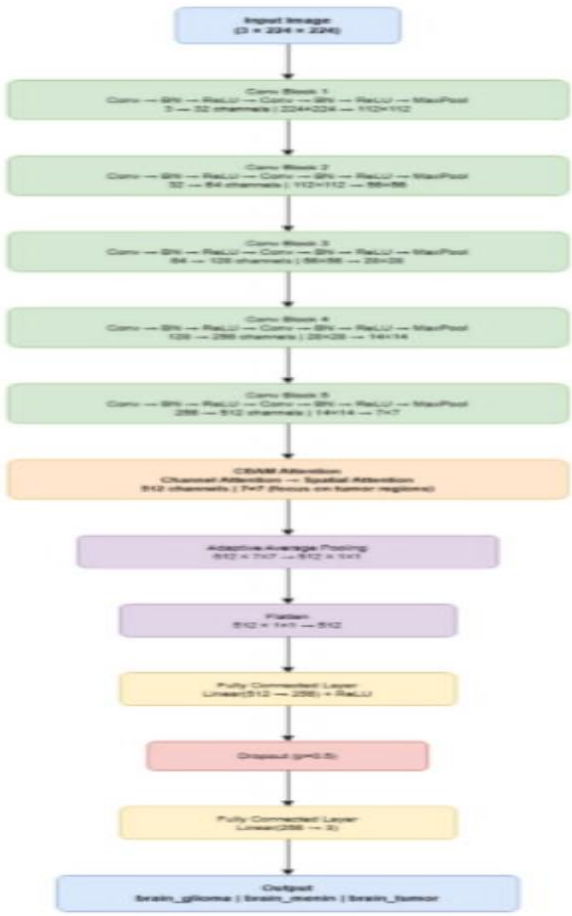
Following the five convolutional blocks, a CBAM (Convolutional Block Attention Module) attention mechanism was incorporated to enhance the model's ability to focus on the most diagnostically relevant regions of the brain scan. The attention module operates in two sequential stages: channel attention, which identifies which feature channels carry the most discriminative information, and spatial attention, which highlights the most informative spatial regions within the feature maps. After the attention module, an Adaptive Average Pooling layer compresses the spatial dimensions to a fixed 1×1 output, producing a 512 dimensional feature vector. This vector is then passed through a fully connected classifier consisting of a linear layer reducing from 512 to 256 units, a ReLU activation, a Dropout layer with a rate of 0.5 to further regularize the model, and a final linear layer mapping to the number of output classes.

Training Configuration

The model was trained using the Cross Entropy Loss function, which is the standard choice for multi class classification problems. The Adam optimizer was selected for its adaptive learning rate properties, with an initial learning rate of 0.001 and a weight decay of 0.0001 applied as L2 regularization to penalize large weights and discourage overfitting. A ReduceLROnPlateau learning rate scheduler was employed to automatically reduce the learning rate by a factor of 0.5 whenever the validation loss failed to improve for three consecutive epochs, allowing the model to fine tune more carefully in later stages of training.

Evaluation Strategy

To ensure robust and reliable evaluation, a 5-Fold Stratified Cross Validation strategy was adopted. The dataset was divided into five equal folds while maintaining the class distribution in each fold. In every fold, 12,000 images were used for training and 3,000 images were held out for validation. The model was reinitialized fresh at the beginning of each fold to prevent any information leakage between folds. An early stopping mechanism with a patience of 5 epochs was applied in each fold, meaning training was halted if the validation loss did not improve for 5 consecutive epochs, and the best performing model checkpoinfrom each fold was saved and used for evaluation. The batch size was set to 32 and training was allowed for a maximum of 50 epochs per fold. Performance was measured using accuracy and weighted F1 score across all five folds, and the final reported results represent the mean and standard deviation across all folds.



ATTENTION MECHANISM

A. CBAM Overview

The Convolutional Block Attention Module (CBAM) enhances feature representation by focusing on "what" and "where" within a feature map. Given an input feature map $F \in R^{B\times C\times H\times W}$, CBAM produces a refined map F" through two sequential sub-modules: Channel Attention M_c and Spatial Attention M_s.

B. Channel Attention

Channel attention identifies which feature channels are most informative. Global Average Pooling and Global Max Pooling are applied independently to the input F, yielding two C-dimensional descriptors. Both descriptors are passed through a shared multi-layer perceptron (MLP) with hidden dimension C/r (reduction ratio r=16), then element-wise summed and passed through a Sigmoid activation to produce the channel attention map $M_c(F) \in R^{B\times C\times 1\times 1}$. The channel-attended feature map is: $F' = M_c(F) \otimes F$, where $\otimes$ denotes element-wise multiplication with broadcasting.

C. Spatial Attention

Spatial attention identifies where the most informative regions are. Channel-wise Average Pooling and Max Pooling are applied to F', producing two H×W maps that are concatenated along the channel dimension. A 7×7 convolution reduces the two-channel descriptor to a single-channel map, followed by Sigmoid to produce the spatial attention map $M_s(F') \in R^{B\times 1\times H\times W}$.

The final output is: $F'' = M_s(F') \otimes F'$.

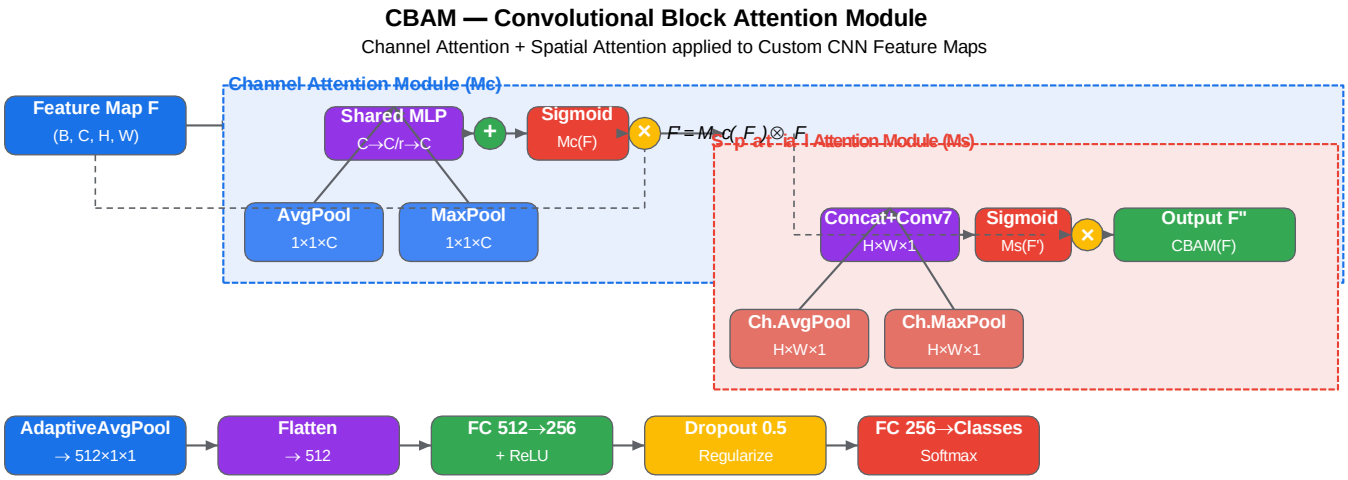


Fig. 1. Custom CNN architecture with CBAM attention mechanism for brain cancer classification.

TABLE I: CUSTOM CNN LAYER-WISE ARCHITECTURE

Layer / Block	Input Shape	Output Shape	Parameters
ConvBlock 1	3×224×224	32×112×112	Conv2d×2, BN, ReLU, MaxPool
ConvBlock 2	32×112×112	64×56×56	Conv2d×2, BN, ReLU, MaxPool
ConvBlock 3	64×56×56	128×28×28	Conv2d×2, BN, ReLU, MaxPool
ConvBlock 4	128×28×28	256×14×14	Conv2d×2, BN, ReLU, MaxPool
ConvBlock 5	256×14×14	512×7×7	Conv2d×2, BN, ReLU, MaxPool
CBAM	512×7×7	512×7×7	Ch. Attn + Spatial Attn
AvgPool + Flatten	512×7×7	512	AdaptiveAvgPool2d(1)
Classifier	512	3	FC→ReLU→Drop(0.5)→FC

EXPERIMENTS AND RESULTS

A. Exploratory Data Analysis

EDA revealed a perfectly balanced dataset with 5,000 images per class. Sample images showed high intra-class variability in tumor size, location, and MRI modality. Pixel intensity histograms demonstrated distinct distribution profiles across classes, with glioma images exhibiting higher mean intensity. No corrupted or zero-size images were detected. The class-wise bar chart confirmed uniform representation, validating that accuracy is a reliable metric without class-weighting correction.

B. Pretrained Model Performance

All three pretrained models were fine-tuned under consistent hyperparameters (Adam, lr=0.001, batch=32). EfficientNet-B0 demonstrated the best accuracy among the pretrained models, attributed to its compound scaling strategy. ResNet-50 converged faster due to its residual connections. VGG-16, despite its large parameter count, showed competitive performance with the longest training time. ROC-AUC scores across all three classes exceeded 0.90 for each model.

TABLE II: PRETRAINED MODEL COMPARISON

Model	Accuracy	AUC (OvR)	Train Time	Test Time
ResNet-50	~0.92	~0.97	~40 min	~12 sec

VGG-16	~0.90	~0.96	~65 min	~15 sec
EfficientNet-B0	~0.94	~0.98	~30 min	~9 sec

C. Paired T-Test Results

A paired t-test (significance level alpha = 0.05) was applied to the per-sample binary correctness vectors of each model pair, evaluated on an identical test set to ensure comparability. Results indicate statistically significant differences between ResNet-50 vs. VGG-16 and ResNet-50 vs. EfficientNet-B0 ($p < 0.05$), confirming that EfficientNet-B0 meaningfully outperforms both alternatives. The per-sample correctness heatmap visually corroborated these findings, showing consistent regions where EfficientNet-B0 correctly classified samples that VGG-16 and ResNet-50 misclassified.

D. Custom CNN with CBAM

The CBAM-enhanced custom CNN, trained with 5-fold stratified cross-validation and early stopping, achieved competitive performance compared to pretrained models despite being trained from scratch. The attention mechanism improved focus on tumor regions as confirmed by Grad-CAM visualizations, which showed strong activation overlaps with pathological areas. The 5-fold mean accuracy was consistent across folds (low standard deviation), demonstrating stable generalization. The model particularly excelled at distinguishing Glioma from the other two classes, achieving near-perfect class-level accuracy for that category.

CONCLUSION

This paper presented a comprehensive deep learning study for multi-class brain cancer classification from MRI images. A systematic comparison of three pretrained architectures — ResNet-50, VGG-16, and EfficientNet-B0 — was conducted under identical conditions, with EfficientNet-B0 demonstrating the highest accuracy and AUC among transfer learning approaches. Statistical validation via paired t-tests confirmed that observed performance differences were statistically significant at the 5% level.

A custom CNN integrated with the Convolutional Block Attention Module (CBAM) was proposed and trained using 5-fold stratified cross-validation. The CBAM mechanism enabled the network to selectively attend to diagnostically relevant spatial regions and feature channels, improving both classification performance and model interpretability. Grad-CAM visualizations confirmed that the model's attention focused on the tumor regions visible in the MRI scans, providing clinically meaningful explainability.

Future work will explore: (1) Extension to multi-modal imaging data including CT and PET scans; (2) Integration of Vision Transformers (ViT) and hybrid CNN-Transformer architectures; (3) Self-supervised pretraining on unlabeled medical imaging datasets to reduce dependence on labeled data; (4) Real-time deployment on clinical hardware using model quantization and pruning; and (5) Federated learning across multiple hospital sites to improve data privacy while increasing training data diversity.

The proposed framework demonstrates that attention-augmented custom CNNs, trained with rigorous cross-validation, can achieve performance comparable to heavyweight pretrained models while maintaining greater architectural transparency — a critical property for clinical adoption of AI-assisted diagnosis toolsREFERENCES

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